

LINEAR SEARCH

```

import time

def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1

n = int(input("Enter the number of elements: "))

arr = []
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))

key = int(input("Enter the element to search: "))

start = time.time()

result = linear_search(arr, key)

end = time.time()

if result != -1:
    print("Element found at index:", result)
else:
    print("Element not found")

print("Time Complexity: O(n)")
print("Execution Time:", end - start, "seconds")

```

```

Enter the number of elements: 5
Enter the elements:
10
20
40
60
80
Enter the element to search: 40
Element found at index: 2
Time Complexity: O(n)
Execution Time: 8.726119995117188e-05 seconds

```

BINARY SEARCH

```

import time

def binary_search(arr, key):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return -1

n = int(input("Enter the number of elements: "))

arr = []
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))

arr.sort()

```

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print("Sorted List:", arr)

key = int(input("Enter the element to search: "))

start = time.time()

result = binary_search(arr, key)

end = time.time()

if result != -1:
    print("Element found at index:", result)
else:
    print("Element not found")

print("Time Complexity: O(log n)")
print("Execution Time:", end - start, "seconds")
```

```
Enter the number of elements: 5
Enter the elements:
10
30
60
20
40
Sorted List: [10, 20, 30, 40, 60]
Enter the element to search: 20
Element found at index: 1
Time Complexity: O(log n)
Execution Time: 8.821487426757812e-05 seconds
```